

# A Proof of the Partial Converse to the Cauchy - Riemann Equations

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**Theorem 1.** *Let  $f(z) = u(x, y) + i(v(x, y))$  be a complex valued function on a domain  $\Omega$  (recall a domain is an open connected set). Now suppose  $u$  and  $v$  has continuous partial derivatives and satisfy the Cauchy - Riemann equations. Then  $f$  will be holomorphic in  $\Omega$ .*

*Proof.* Let  $z_0 = x_0 + iy_0 \in \Omega$  be arbitrary where  $x_0$  and  $y_0$  are real numbers. Since  $\Omega$  is open,  $\exists r > 0$  such that  $D(z_0, r) \subset \Omega$ .

Take  $h = p + qi$  such that  $|h| < r$  and  $p$  and  $q$  are real numbers. Then we will try to show that the following limit, call it  $L$  exists.

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h} &= \frac{u(x + p, y + q) - u(x, y) + i(v(x + p, y + q) - v(x, y))}{h} \\ &= \frac{u(x + p, y + q) - u(x, y + q) + i(v(x + p, y + q) - v(x, y + q))}{h} \\ &\quad + \frac{u(x, y + q) - u(x, y) + i(v(x, y + q) - v(x, y))}{h} \\ &= \frac{p}{h} \left( \frac{u(x + p, y + q) - u(x, y + q) + i(v(x + p, y + q) - v(x, y + q))}{p} \right) \\ &\quad + \frac{q}{h} \left( \frac{u(x, y + q) - u(x, y) + i(v(x, y + q) - v(x, y))}{q} \right) \\ &= \frac{p}{h} \left( \frac{u(x + p, y + q) - u(x, y + q)}{p} \right) + i \frac{p}{h} \left( \frac{v(x + p, y + q) - v(x, y + q)}{p} \right) \\ &\quad + \frac{q}{h} \left( \frac{u(x, y + q) - u(x, y)}{q} \right) + i \frac{q}{h} \left( \frac{v(x, y + q) - v(x, y)}{q} \right) \end{aligned}$$

Notice that in the second line we add and subtract  $u(x, y + q)$  and  $iv(x, y + q)$

We now apply the mean-value theorem to each of the 4 quotients given above. This is valid, because we know  $u$  and  $v$  are continuous and differentiable.

Hence  $\exists \alpha, \beta, \gamma, \delta \in (0, 1)$  for  $i = 1, 2, 3, 4$  such that our quotient from above:

$$= \frac{p}{h} \frac{\partial u}{\partial x}(x + \alpha p, y + q) + i \frac{p}{h} \frac{\partial v}{\partial x}(x + \beta p, y + q) + \frac{q}{h} \frac{\partial u}{\partial y}(x, y + \gamma q) + i \frac{q}{h} \frac{\partial v}{\partial y}(x, y + \delta q)$$

Now we use the Cauchy-Riemann equations  $u_x = v_y$  and  $u_y = -v_x$  to write the above expression in terms of  $u_x$  and  $v_x$ . Then our quotient will be:

$$= \frac{p}{h} \frac{\partial u}{\partial x}(x + \alpha p, y + q) + i \frac{q}{h} \frac{\partial u}{\partial x}(x, y + \delta q) + i \frac{p}{h} \frac{\partial v}{\partial x}(x + \beta p, y + q) - \frac{q}{h} \frac{\partial v}{\partial x}(x, y + \gamma q)$$

Now we rewrite the above expression in an attempt to demonstrate that its limit as  $h$  tends to 0 will be  $u_x(x, y) + iv_x(x, y)$ . First we define the following new expressions

$$U(x, y) = \frac{p}{h} \frac{\partial u}{\partial x}(x + \alpha p, y + q) + i \frac{q}{h} \frac{\partial u}{\partial x}(x, y + \delta q)$$

$$V(x, y) = i \frac{p}{h} \frac{\partial v}{\partial x}(x + \beta p, y + q) - \frac{q}{h} \frac{\partial v}{\partial x}(x, y + \gamma q)$$

We now show that  $U(x, y)$  tends to  $u_x(x, y)$  and  $V(x, y)$  tends to  $iv_x(x, y)$  as  $h$  tends to zero. We first rewrite  $U(x, y)$  as  $u_x(x, y)$  and two error terms that will subsequently tend to zero as  $h$  tends to zero:

$$U(x, y) = u_x(x, y) + E_1(x, y) + E_2(x, y)$$

$$E_1(x, y) = \frac{p}{h} \frac{\partial u}{\partial x}(x + \alpha p, y + q) - \frac{p}{h} u_x(x, y)$$

$$E_2(x, y) = i \frac{q}{h} \frac{\partial u}{\partial x}(x, y + \delta q) - i \frac{q}{h} u_x(x, y)$$

However we have the following bounds on  $p, q$  and  $h$ .

$$\left| \frac{p}{h} \right| \leq 1$$

$$\left| \frac{q}{h} \right| \leq 1$$

Furthermore, by continuity of the first partial derivatives, we get

$$\lim_{h \rightarrow 0} u_x(x + \alpha p, y + q) = u_x(x, y)$$

$$\lim_{h \rightarrow 0} u_x(x, y + \delta q) = u_x(x, y)$$

Then we use the sandwich theorem, as follows:

$$0 \leq |E_1| = \left| \frac{p}{h} \right| |u_x(x + \alpha p, y + q) - u_x(x, y)| \leq |u_x(x + \alpha p, y + q) - u_x(x, y)|$$

And taking the limit as  $h$  tends to zero we find that  $E_1$  tends to zero. Analogously, it can be shown that  $E_2$  tends to zero and by rewriting  $V(x, y)$  as shown below with errors  $E_3$  and  $E_4$ , it can similarly be shown that  $V(x, y)$  tends to  $iv_x(x, y)$  as  $h$  tends to zero by using continuity and the sandwich theorem.

$$\begin{aligned} V(x, y) &= iv_x(x, y) + E_3(x, y) + E_4(x, y) \\ E_3(x, y) &= i \frac{p}{h} \frac{\partial v}{\partial x}(x + \beta p, y + q) - i \frac{p}{h} v_x(x, y) \\ E_4(x, y) &= -\frac{q}{h} \frac{\partial v}{\partial x}(x, y + \gamma q) + \frac{q}{h} v_x(x, y) \end{aligned}$$

Thus the limit  $L$  exists and its value is shown below

$$\lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h} = u_x(x, y) + iv_x(x, y)$$

□